We have already learnt how to find the perimeter and the area of shapes like circles, triangles, squares, rectangles, trapeziums, parallelograms etc. In this chapter, we shall see the parts of a circle and how to find the perimeter and the area of a sector and some combined shapes.

Recap

The teacher asks the students to calculate the area of the circle of radius 7 *cm*. Many students have solved it as in Method 1, but a few of them have done it as in Method 2.

Method 1:	**Method 2:**
Area of the circle, A $= \pi r^2$ *sq.units*	Area of the circle, A $= \pi r^2$ *sq.units*
$= \frac{22}{7} \times 7 \times 7$	$= 3.14 \times 7 \times 7$
$= 154$ *sq.cm*	$= 153.86$ *sq.cm*

Then, they had the following conversation:

Sudha : Which of these two answers is correct, Teacher ?

Teacher : Both answers are correct but only approximate.

Sudha : We get two answers for the same question. How is it possible, teacher?

Teacher : We can infact solve this exactly as $\pi \times 7 \times 7 = 49\pi$ *sq.cm.*

Meena : When we find the area of a square, a rectangle etc., we all get unique answers, but why is the area of a circle not unique ?

Teacher : What is the value of π ?

Meena : The value of π is $\frac{22}{7}$ or 3.14

Teacher : Actually it is neither $\frac{22}{7}$ nor 3.14. They are only approximate values.

Meena : Then, what is the exact value of π?

Teacher : Though π is the constant, it is a non-terminating and non-recurring decimal number. So, we are not able to use its exact value and to make the calculations easy, $\frac{22}{7}$ or 3.14 is used as the approximate value of π.

Meena : Is the area of a circle always approximate, teacher?

Teacher : No, it is an exact answer unless we substitute the value of π as $\frac{22}{7}$ or 3.14.

Meena : Then, for the above problem, 49π *sq.cm* is an exact answer whereas 154 *sq.cm* and 153.86 *sq.cm.* are approximate answers. Am I right, teacher?

Teacher : Yes, you are right Meena.

Think



1. $\frac{22}{7}$ and 3.14 are rational numbers. Is 'π' a rational number? Why?
2. When is the 'π' day celebrated? Why?

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